MILITARY SPECIFICATION

SERVICE UNIT, REFRIGERATION SYSTEM (MUST)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a service unit for the air conditioner refrigeration system of the Power Plant, Utility (MUST).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

J-C-175	- Cable Assembly, Power, Electrical, (3-Wire, 3-Prong, Grounding Plug Connector, for 125-Volt Equipment).
J-C-580	- Cord, Flexible, and Wire, Fixture, (Electrical, 0- to 600-Volt Service).
W-C-596	- Connector, Plug, Electrical; Connector, Receptacle, Electrical.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
QQ-B-654	- Brazing Alloys, Silver.
QQ-S-571	- Solder; Tin Alloy; Lead-Tin Alloy and Lead Alloy.
QQ-S-698	- Steel, Sheet and Strip, Low-Carbon.
QQ-S-700	- Steel Sheet and Strip, Medium and High Carbon.
QQ-S-781	- Strapping, Steel, and Seals.
WW-T-775	- Tube, Copper, Seamless, (for Refrigeration and General Use).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

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Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-W-5086/1	- Wire, Electric, Polyvinyl Chloride Insulated, Nylon Jacket, Tin-Coated Copper Conductor, 600-Volt, 105° C.
MIL-F-18866	- Fittings, Hydraulic Tube, Flared, 37 Degree and Flareless; Steel.
MIL-M-43719	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented; General Specification for.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-882	- System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for.
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment and Facilities.
MS16658	- Connector, Plug and Duplex Receptacle, Electrical, Power, 2-Pole, 3-Wire, Grounding Type, 15 Amperes, 125 Volts.
MS25036	- Terminal, Lug, Crimp Style, Copper, Insulated, Ring - Tongue, Bell-Mouthed, Type II, Class 1.
MS35793	- Handle, Bail-Chest, Surface Mounting, Plain Holes.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B70.1 - Refrigerant Flare Type Fittings.
B9.1 - Safety Code for Mechanical Refrigeration.

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code, Refrigeration Flare Type Fittings.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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3. REQUIREMENTS

3.1 Description. The service unit shall consist of a vacuum system, refrigerant charging system, oil charging system, refrigerant gages, and necessary servicing accessories; all contained in a portable skid-mounted inclosure. The components, parts, and interconnecting piping shall be as shown in Figure 1 and as specified herein. The maximum dry weight of the service unit with accessory equipment shall be 300 pounds.

3.2 First article (preproduction model). The supplier shall furnish one or more service units as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce service units that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more service units as specified (see 6.2), for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Materials shall be new and free from defects or imperfections that may affect the serviceability or appearance of the finished service unit. Materials subject to atmospheric corrosion or rust shall not be treated with a temporary treatment such as grease or oil in order to meet environmental requirements.

3.4.1 Fasteners. All steel fasteners shall be cadmium, zinc, or nickel plated to resist atmospheric corrosion under the environmental conditions specified herein. Sheet metal or self-tapping screws shall not be used. Threads shall conform to Handbook H28.

3.4.2 Tube, copper. Copper tubing shall conform to WW-T-775.

3.4.3 Joining material.

3.4.3.1 Solder. Solder shall conform to QQ-S-571, Type Designation Sb5 WS.

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3.4.3.2 Brazing. Brazing material shall conform to QQ-B-654, Grade 4 or 6.

3.4.4 Sheet and strip steel. Sheet and strip steel shall conform to QQ-S-698 or QQ-S-700.

3.4.5 Aluminum alloy. The aluminum alloy sheet or plate shall conform to QQ-A-250/11.

3.5 Environmental conditions.

3.5.1 Operating temperatures. The service unit shall be capable of operating in any ambient temperature from plus 50°F to 120°F.

3.5.2 Storage. The service unit shall not be damaged and its operation shall not be impaired by storage at an ambient temperature of plus 160°F for 4 hours each 24 hours, no solar radiation, and at minus 65°F for 6 hours each 24 hours, no solar radiation.

3.5.3 Humidity. After being subjected to the test specified in 4.6.3.7, there shall be no evidence of water or corrosion inside the enclosure and the service unit shall start and operate without malfunction, electrical shorts, or change in tolerance limits of any internal or external parts which could prevent the unit from meeting operational requirements.

3.5.4 Shock. When tested as specified in 4.6.3.5, the service unit shall start and operate without malfunction and there shall be no evidence of fracture or breakdown of any material.

3.5.5 Rain. After being subjected to the test specified in 4.6.3.11, there shall be no evidence of water or corrosion inside the enclosure and the service unit shall start and operate without malfunction, electric shorts, or change in tolerance limits of any internal or external parts which could prevent the unit from meeting operational requirements.

3.6 Safety. All belts, rotating parts, and moving parts shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. Sight level gages such as used with the refrigerant metering tube (see 3.9.5.1) and the oil measuring tube (see 3.9.6.1) shall have a protective shield to guard personnel from the potential hazard or rupture. The vent openings shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazards. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. The refrigerant charging system (see 3.9.5) shall conform to the applicable requirements of ANSI B9.1.

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3.7 Human factors. The service unit shall comply with the applicable requirements of MIL-STD-1472. The piping and components of the vacuum system refrigerant charging system, and oil metering system shall be arranged so that the systems are visually traceable from the front of the unit and when the unit is being operated. The piping shall be straight as possible, and the components of each system shall be grouped so that the lines do not cross. All valve handles shall be accessible from and face the front of the unit. The piping of each system shall also be color coded (see 3.12). All valves, connectors, scales, switches, and other items which are used by the operator shall be labeled in accordance with MIL-STD-1472. Labels shall be legible from 6 feet. The accessibility of all connectors shall be such that all accessory hoses may be attached to the service unit with a standard open end wrench and each connector shall be snubbed to prevent turning when attaching and removing hoses.

3.8 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement or maintenance shall be equipped with quick-disconnect fastening. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.9 Technical requirements.

3.9.1 Piping. All piping shall be copper tubing. Copper tubing shall conform to the material specified in 3.4.2. Fittings shall be wrought copper solder type or where required, refrigeration flare type fittings conforming to ANSI B70.1 and bulkhead fittings conforming to MIL-F-18866. Hand valves shall be refrigerant diaphram packless type. Tubing connections (where flare type fittings are not specified) shall be brazed using material specified in 3.4.3.2 and shall be used without flux. Where brazing temperatures will have a deleterious effect on a fitting or component, the joint shall be soldered using material specified in 3.4.3.1. All tubing, fittings, and valves shall be supported and located to prevent abrasive action with structural members, other components, and other tubing. All terminal connections used for attaching service hoses shall be flare type and shall have removable seal caps with a keeper chain.

3.9.2 Wiring. All electrical wire shall be in accordance with MIL-W-5086/1. Wire shall be of adequate size for the application and shall be installed in accordance with NFPA No. 70. All wire shall be secured in compack harnesses and attached to frame members with insulated clamps at close intervals to insure a neat and orderly cable run. Wiring shall not be attached to panels or other components which may be removed during maintenance. Electrical isolators shall be used between all wiring and metal components to prevent abrasion action to wire insulation. All

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wiring connections, including screw types such as terminal block, shall have protective covers to prevent exposed connections, electrical shorts, and shock to personnel. The ends of all wire which terminates at a screw-type connector shall be equipped with terminals conforming to MS25036. All wiring passing through a panel shall be grommeted.

3.9.3 Enclosure. All components and accessories of the service unit shall be contained in a rigid and sealed enclosure. The enclosure shall be fabricated in two mating sections which are hinged together. The joint between the two sections shall be gasketed to provide a waterproof seal when drawn together with fasteners. The service unit shall be ready to use when the sections are opened.

3.9.3.1 Size of material. The maximum exterior dimensions of the enclosure are 26 inches deep by 31 inches wide by 42 inches high. The enclosure shall be fabricated with aluminum or steel conforming to 3.4.

3.9.3.2 Interior. The vacuum system (3.9.4), refrigerant charging system (3.9.5), oil charging system (3.9.6), electrical system (3.9.7), and pressure gages (3.9.8) shall be permanently and securely mounted inside the enclosure. Compartments, containers, and securing means shall also be provided inside the inclosure for the stowage of all accessory equipment (3.9.9).

3.9.3.3 Fasteners. Fasteners for drawing the two sections together shall be permanently attached to the enclosure and shall be the spring-loaded cam type with a wing nut type handle that turns 180 degrees to open or lock. The fastener shall be capable of compressing the gasket to insure a watertight seal and shall provide takeup for set in the gasket.

3.9.3.4 Lifting handles. Four lifting handles conforming to MS35793 shall be provided for moving the unit. The handles shall be spaced so that the unit may be carried by four men.

3.9.3.5 Air pressure equalizer valve. A manually operated valve shall be provided to equalize the pressure inside the sealed enclosure during air transportation.

3.9.3.6 Skids. The enclosure shall be mounted on skids 3 inches high. The enclosure in either the open or closed position shall be stable when setting on the skids.

3.9.4 Vacuum system. The vacuum system shall consist of a vacuum pump, electronic high vacuum gage, interconnecting piping, and fittings and shall be arranged in accordance with schematic diagram, Figure 1. When the

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refrigeration service unit is in a level position, drainage of vacuum pump oil shall be to the ground without draining on or over any part of the unit. Accessibility to the oil fill opening, the oil level indicator or drain shall be provided without the removal or adjustment of parts. Accessory equipment for drainage and filling is permitted. The system shall produce a vacuum of less than 100 microns mercury (Hg) when tested as specified in 4.6.3.1.2 and 4.6.3.1.4. When tested as specified in 4.6.3.1.3, the system shall produce less than 30 microns Hg.

3.9.4.1 Vacuum pump. The vacuum pump shall have a minimum piston displacement of 3 cubic feet per minute and shall have a minimum "blank off" vacuum of 5 microns of mercury as measured on a McLeod gage. The pump shall be electric motor driven. The motor shall be 115 volts, single phase, 60 Hertz, equipped with thermal protection, and shall be capable of driving the pump continuously. The vacuum pump shall have an oil level indicator.

3.9.4.2 Electronic vacuum gage. The electronic vacuum gage shall have a minimum range of 1 micron to millimeter (mm) mercury total absolute pressure. The scale length shall not be less than 3 inches, shall be expanded logarithmetically with the following minimum scale graduation:

Micron Range	Maximum Equivalency of One Graduation, Microns
10 to 100	5
100 to 200	10
200 to 500	50
500 to 1000	100

The minimum accuracy of the gage shall be as follows:

100 microns	± 15 microns
200 microns	± 25 microns
500 microns and up	± 10 percent of reading

3.9.5 Refrigerant charging system. The refrigerant charging system shall consist of a refrigerant metering tank, electric heater control system over-pressure and over-temperature protection system, and interconnecting tubing and fittings and shall be arranged as shown in Figure 1. When subjected to the test specified in 4.6.3.2.1, the completely assembled system shall withstand pressures up to 100 psig at 145° F without deformation or any detectable leak. When tested in accordance with 4.6.3.2.4 the refrigerant charging system shall charge refrigerant with an accuracy of 95 percent or greater.

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3.9.5.1 Refrigerant metering tank. The refrigerant metering tank shall store and meter a minimum of 90 pounds of liquid Refrigerant-114 at 135° F. The refrigerant shall be metered by measuring the level of refrigerant in the tank with a sight glass and graduate scale. The tank shall have an expansion volume of not less than 10 percent when filled to the maximum metering level.

3.9.5.1.1 Metering tube and scale. The metering tube shall be constructed of a transparent, oil resistant, impact resistant rigid elastomer. The scale shall be calibrated in pounds of Refrigerant-114 at 135° F and shall be graduated so that one division is equivalent to 1 pound or less and shall be figured at 5 pound intervals. The spacing between each division shall be 3/16 inch or less. The entire scale shall be readable when the enclosure door is open.

3.9.5.1.2 Metering tank heater. The metering tank shall be heated with an electric heater to obtain the necessary refrigerant pressure for charging refrigerant. The electric heater shall be of sufficient size to heat a full tank of refrigerant from 60° F to 145° F within 1 hour, when the service unit is in an ambient temperature of 60° to 70° F. Thermal insulation may be used around the tank to lower the electric heater wattage.

3.9.5.2 Electric heater control system. When the refrigerant charging tank contains liquid refrigerant, an automatic control system shall maintain the tank pressure between 58 and 70 psig. If the tank is depleted of liquid refrigerant the tank temperature shall not exceed 145° F. If the automatic control system fails, the electric heaters shall be disconnected to prevent the tank pressure from exceeding 90 psig.

3.9.5.3 Relief valve. The tank relief valve shall be in accordance with ANSI Safety Code for Mechanical Refrigeration, B9.1.

3.9.5.4 Refrigerant drier and fitting. A tee fitting suitable for accomodating a tee flow type drier shall be installed where indicated on Figure 1. The drier shall be designed for refrigerant use, shall have a minimum capacity of 8 cubic inches and shall be equivalent to the ANSUL T-FLO Drier System.

3.9.6 Oil charging system. The oil charging system shall consist of an oil measuring tube, connecting piping and fittings and shall be arranged as shown in Figure 1. When tested as specified in 4.6.3.3.1, the system shall meter oil with an accuracy of not less than 97.5 percent.

3.9.6.1 Oil measuring tube. The measuring tube shall have a measurable capacity of not less than 1000 cubic centimeters (cc). The tube shall be constructed of a transparent, oil resistant, impact resistant, rigid elastomer. The oil level shall be visually determined with a scale which is calibrated in 10 cc graduations. The scale shall be permanently marked on the tube and the marking shall resist removal by slight abrasion. The ends of the tube shall be closed and shall contain a fitting for connecting tubing.

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3.9.7 Electrical system. The electrical system shall operate on 115 volt, single phase, 60 Hertz power and shall provide the following:

(a) Manual selection of the vacuum pump.
(b) Manual selection of the refrigerant metering tank heater.
(c) Automatic control of the metering tank pressure.
(d) Protection against excessive tank pressure and temperature by de-energizing the tank heater.
(e) A power source for the electronic vacuum pressure gage.
(f) A convenience outlet conforming to MS16658.
(g) A power supply cord.
(h) Overcurrent protection.
(i) Grounding of all exposed noncurrent carrying metal parts.

3.9.7.1 Power supply cord. The power supply cord shall conform to J-C-175. One end of the cord shall attach to the refrigeration service unit and the other end shall have a male plug connector. The length of the cord shall be 12 feet. During storage and shipment the power supply cord shall be secured inside the enclosure.

3.9.8 Pressure gages. Pressure gages for measuring the refrigerant high side pressure and refrigerant low side pressures during servicing shall be permanently mounted within the service unit cabinet. Connections to the gages shall be in accordance with Figure 1. Gages shall be 2-1/2 inch dial size with an accuracy of 1 percent of scale range within the middle working half and 2 percent over the remainder of the scale. The high pressure gage shall have a range of 30 inches mercury (Hg) vacuum to 0 to 150 pounds per square inch. The low side gage shall have a range of 30 inches Hg to 0 to 60 pounds per square inch.

3.9.9 Accessory equipment. The following accessory equipment shall be furnished with each service unit:

Quantity	Unit	Description
1	ea	Vacuum hose, 1/2 I.D. X 20 ft. long, end connections 3/8 female flare and 1/2 female flare.
1	ea	Vacuum hose, 1/2 I.D. X 6 ft. long end connections 3/8 female flare and 1/2 female flare.
1	ea	Refrigerant hose, 1/4 in. dia. X 20 ft. long, end connections 1/4 in. female flare and 3/8 in. female flare.

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Quantity	Unit	Description
2	ea	Refrigerant hose 1/4 in. dia. X 5 ft. long, end connections 1/4 in. female flare and 3/8 in. female flare.
1	ea	Refrigerant charging hose 3/8 in. dia. X 5 ft. long, end connections 3/8 in. female flare.
1	ea	Extension cord.
1	ea	Centrifuge tube.
1	ea	Oil pump, manually operated.
1	qt	Vacuum pump oil.
10	ea	Refrigerant hose gaskets, 1/4 in.
10	ea	Refrigerant hose gaskets, 3/8 in.
10	ea	Refrigerant hose gagkets, 1/2 in.

Each accessory item shall be stored in a designated space within the service unit to provide neat and orderly storage. The necessary straps, bins, and other means of holding the accessory items in order to prevent damage during shipment and handling shall be built into the cabinet. All hoses shall be equipped with "quick-coupler" type hose couplings containing replaceable soft composition gaskets. The hose couplings shall require only hand tightening to make a joint that does not have a leakage rate greater than when tested as specified in 4.6.3.2.1. The hose couplings shall mate with male flare fittings conforming to ANSI B70.1. All hose couplings of the same size shall use the same replaceable gasket.

3.9.9.1 Vacuum hoses. Vacuum hoses shall be for high vacuum service and shall be capable of bending on 8 inch radius at 50° F. The vacuum hose shall also qualify when tested as specified in 4.6.3.1.

3.9.9.2 Refrigerant hoses. Hoses used for charging refrigerant and oil, connecting refrigerant pressure gages, and other refrigerant servicing operations shall be the commercially standard type used for this service. The hoses shall have a safe working pressure of 500 psig and a minimum burst pressure of 1000 psig. The hoses shall have a neoprene or Buna-N interior tube and a rayon braid cover with an outer cover of neoprene.

3.9.9.3 Oil pump. The manually operated oil pump shall deliver oil at a minimum rate of 200 mil per minute and at a discharge pressure of 120 psig. The suction and discharge connections shall be 1/4 inch male flare.

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3.9.9.4 Centrifuge tube. The glass pear shaped centrifuge tube for measuring the percentage of oil circulating with the refrigerant shall conform to Kimble Part No. 45245 or equal.

3.9.9.5 Extension cord. The extension cord shall be 25 feet plus or minus 3 inches in length and shall consist of an assembly fabricated from flexible cord, Type SJ or SJO in accordance with J-C-580, a male plug connector, and a female receptacle conforming to W-C-596 Series D. The ampere capacity of the cord as given in the National Electric Code is to be not less than the measured input to the service unit. The connectors shall be of the mechanical cord-grip style or the type that is molded to the cable jacket.

3.10 Identification marking. The refrigeration service unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the refrigeration service unit on plates conforming to MIL-P-514, Type I, Style 3, Comparison C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.11 Instruction decal. The instruction decal shall be in accordance with MIL-M-43719; type and class optional. All lettering shall be black Color No. 17038 in accordance with FED. STD. No. 595 and background shall be insignia white. The instruction decal shall be attached to an inside surface of the service unit and in a location which is readable when in normal operation. The instruction decal shall contain a schematic diagram of the refrigeration service unit in accordance with Figure 1, and operating instructions to perform the following:

(a) Testing for leaks.
(b) Evacuation of the service unit.
(c) Transferring refrigerant to the unit metering tank.
(d) Evacuation of air conditioner refrigeration system.
(e) Charging air conditioner refrigeration system with refrigerant.
(f) Filling the unit oil system.
(g) Charging air conditioner refrigerant system with oil.

3.12 Treatment and painting. The portions of the service unit normally painted shall be cleaned and painted in accordance with MIL-T-704, Type A. All colors shall conform to FED. STD. No. 595. The color of the inside and outside of the enclosure, and skids shall be olive drab lusterless No. 34087. Each system of the refrigeration service unit shall be color coded for tracing and operating as follows:

System	Color	FED. STD. No. 595
(a) Refrigerant Charging System	orange	12473
(b) Oil Charging System	yellow	13538
(c) Vacuum System	green	14062
(d) Refrigerant High Pressure Gage	red	11105
(e) Refrigerant Low Pressure Gage	blue	15123

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3.13 Workmanship. All parts, components, and assemblies of the service unit including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.13.1 Metal fabrication. Sheet metal and structural members used in fabrication shall be straight and free from laminations. Sheet metal and structural members may be cut by shearing or sawing. All burrs and rough edges shall be removed. Corners shall be square and true. All bends shall be made to a redius which will not cause cracking at the extreme fibers of the piece being bent, and all bends except for minor details shall be made by use of metal dies or fixtures to insure uniformity in size and shape.

3.13.2 Bolted connections. Boltholes shall be punched or drilled and shall have the burrs removed. Washers and lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued.

3.13.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the members.

3.13.4 Jigs and fixtures. Shop-fabricated components shall be assembled in steel jigs or frames and joined while held in position. Jigs and frames shall be designed to minimize distortion.

3.13.5 Interchangeability. All like components shall be interchangeable with respect to installation, performance, and quality.

3.13.6 Cleanliness. Extreme care shall be exercised during all phases of fabrication and assembly to eliminate contaminants such as residual solvents, moisture, metallic particles, flux, dirt, scale, and rust from all systems. This care will include, but shall not be limited to:

(1) Thoroughly cleaning all tubing with Dupon Freon Precision Cleaning Agent or equivalent following fabrication;
(2) the thorough cleaning and degreasing of all components;
(3) sealing all tubing and other components with shipping caps to prevent the entrance of contaminants after cleaning and the removing of their caps only at installation; and
(4) maintaining sealed systems during fabrication whenever possible.

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3.13.7 Brazing. During the brazing operation a small amount of gas such as dry nitrogen shall be continuously bled through the refrigerant lines to prevent internal oxide scaling. Care shall be exercised during the brazing operation to protect all components from deformation and damage. The quantity of brazing alloy and flux used shall be no more than that required to make the joint. Flux shall not be present in the refrigerant system, and heavy beads and fillets of braze material shall be avoided.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction service unit(s) shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction service unit(s) shall be subjected to the tests marked "X" in Column 2 of Table I in the sequence listed or in the sequence listed in a Government approved test plan. Failure of any test shall be cause for rejection.

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4.4 Initial production inspection. When specified (see 3.3), one or more initial production service units will be selected at random by the Government from service units being produced by production tooling and will be subjected to the examination and tests marked "X" in Column 4 of Table I, to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production service unit shall not exclude the remaining service units from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production service unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production service unit(s) and shall be cause for refusal by the Government to continue acceptance of production service units until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on service units previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all service units accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each service unit shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each service unit shall be subjected to the tests marked "X" in Column 3 of Table I in the sequence listed. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The service units shall be examined as specified herein for the following defects:

101. Fasteners not as specified.
102. Stowed items missing or not secured.

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103. Service unit not in accordance with Figure 1.
104. Enclosure dimensions exceed the maximum specified.
105. Mounting skids not as specified.
106. Enclosure handles not as specified.
107. Enclosure sections not hinged as specified.
108. Hinged sections that do not open and close and do not fasten as intended.
109. Any fasteners that do not function as intended.
110. Instruction decal missing or not as specified.
111. Labeling not as specified.
112. Arrangement of system not as specified.
113. Tubing and fitting sizes not as specified.
114. Tubing and components not supported to prevent abrasion with structural member, other components or other tubing.
115. Workmanship not as specified.
116. Weight exceeding the maximum specified.
117. Materials not as specified.
118. Safety provisions not as specified.
119. Wiring and electrical system not as specified.
120. Air pressure equalizer valve missing or not as specified.
121. Gages not as specified.
122. Power supply cord not as specified.
123. Accessory equipment missing, not stored securely or not as specified.
124. Treatment and painting not as specified.

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4.6.2 Inspection schedule. The inspection schedule is listed in Table I.

Table I. Inspection Schedule

Inspec. Sequence	Type of inspection: Pre-prod.	Qual. Conf.	Init. Prod.	Examination or test	Inspec. para.	Requirement para.
1	2	3	4	5	6	7
1	X	X	X	Examination.	4.6.1	3.1, 3.4, 3.9.1, 3.9.2, 3.9.3, 3.9.7, 3.9.9, 3.10, 3.12, 3.13
2	X	-	-	Vacuum system test.	4.6.3.1	3.9.4
3	X	-	-	Refrigerant charging system test.	4.6.3.2	3.9.5
4	X	-	-	Oil charging system test.	4.6.3.3	3.9.6
5	X	-	-	Refrigeration gage calibration test.	4.6.3.4	3.9.8
6	X	-	-	Shock test.	4.6.3.5	3.5.4
7	X	-	-	High temperature.	4.6.3.12	3.5.1, 3.5.2
8	X	-	-	Low temperature storage test.	4.6.3.6	3.5.2
9	X	-	-	Rain.	4.6.3.11	3.5.5
10	X	-	-	Humidity test.	4.6.3.7	3.5.3
11	-	X	X	Service unit check test.	4.6.3.8	3.9.4, 3.9.5 3.9.6
12	-	-	X	Human factors test.	4.6.3.9	3.7
13	-	-	X	Maintenance evaluation.	4.6.3.10	3.8
14	-	-	X	Safety.	4.6.3.13	3.6

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4.6.3 Tests.

4.6.3.1 Vacuum system.

4.6.3.1.1 Electronic vacuum gage. Determine the accuracy of the electronic vacuum gage with a master gage which was calibrated with a McLeod gage. Determine the accuracy at 100, 200, 500, and 1000 microns Hg absolute pressure. Nonconformance to 3.9.4.2 shall constitute failure of this test.

4.6.3.1.2 Vacuum system leakage. Install vacuum hoses on both vacuum connections and connect the ends of the hoses with a flare union. Open valves Vac-1 and Vac-2 (Figure 1) and start the vacuum pump. Operate the pump for 15 minutes and measure the pressure within the system utilizing the electronic vacuum gage. Nonconformance to 3.9.4 shall constitute failure of this test.

4.6.3.1.3 Vacuum pump - "blank-off" pressure. Close valve Vac-1, open valve Vac-2 (Figure 1) and operate vacuum pump. Measure the pressure developed utilizing the electronic vacuum gage. Nonconformance to 3.9.4 shall constitute failure of this test.

4.6.3.1.4 Vacuum system operation. Install vacuum hoses to the service unit and the other ends of the hoses to a tank having a minimum volume of 2 cubic feet. Following the procedures of the instruction decal, evacuate the tank. Evacuation shall continue for 2 hours in an ambient temperature of 80° F to 110° F. Every half hour the vacuum shall be broken, allowing the pressure to return to atmospheric pressure and then evacuated. Before breaking the vacuum, record the following data:

(a) System pressure with the electronic vacuum meter.
(b) Ambient temperature.
(c) Vacuum pump motor terminal voltage.
(d) Vacuum pump motor amperage.

One or more of the following shall constitute failure of the test:

(a) Nonconformance to 3.9.4.
(b) A motor amperage greater than the motor manufacturer's nameplate.
(c) Vacuum system does not operate continuously for 2 hours.

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4.6.3.2 Refrigerant charging system.

4.6.3.2.1 Refrigerant charging system pressure. Test pressure tolerance shall be plus or minus 5 psig. Pressurize the system including the refrigerant hoses to 100 psig with dry nitrogen containing refrigerant gas detectable with a halogen-sensitive electronic leak detector (General Electric Type H-2, or equal). Calibrate the leak detector for the type of gas and percentage of mixture employed, and set the detector to indicate a leakage rate of 0.1 ounce per year. Check the entire refrigerant system for leaks. One or more of the following shall constitute failure of this test:

(a) Any detectable leak.
(b) Any evidence of distortion or deformation.

4.6.3.2.2 Metering tank automatic control system. Place the service unit in a 60° to 70° F ambient temperature and fill the tank to approximately 50 percent of the maximum metering level with R-114 refrigerant. Turn on the tank heater and allow the heater to cycle on and off. Record the high and low tank pressure for three complete cycles. Repeat this test with the tank 10 percent full. Nonconformance to 3.9.5.2 shall constitute failure of this test.

4.6.3.2.3 Safety protection. Fill the refrigerant metering tank approximately 50 percent full of R-114 refrigerant and electrically shunt out the automatic control. Turn on the electric tank heater. Record the maximum tank pressure attained. Repeat this test two times (Note: Do not allow the tank pressure to exceed 100 psig). Nonconformance to 3.9.5.2 shall constitute failure of this test.

4.6.3.2.4 Refrigerant charging system operation. Following the procedure of the instruction decal, evacuate and fill the refrigerant charging system to the maximum level with R-114. Allow the tank to heat and determine if the heater is cycling on and off with the operating control. After the heater has completed two "on-off" cycles, fill an external tank with R-114 in 10-pound ± 1-pound increments using the metering tank scale until the metering tank is empty. Each increment of refrigerant added to the external tank shall be weighed using a scale having an accuracy of not less than 98 percent. Each metering tank increment shall be compared to the weight added to the external tank. After the metering tank is empty, the heater shall remain in the "on" position for at least 1/2 hour. Nonconformance to 3.9.5, 3.9.5.1, 3.9.5.1.1, 3.9.5.1.2, or 3.9.5.8 shall constitute failure of this test.

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4.6.3.3 Oil charging system.

4.6.3.3.1 Oil charging system operation. Following the procedures of the instruction decal, evacuate and fill the oil charging system with refrigerant oil. Using the oil measuring tube scale, measure sucessive quantities of 450 cc, 250 cc and 50 cc into calibrated graduates which have a maximum graduation marking of 5 cc. Nonconformance to 3.9.6 shall constitute failure of this test.

4.6.3.3.2 Oil pump. Following the procedures of the instruction decal, fill the oil charging system and connect the manual oil pump to the service unit. The discharge of the pump shall be connected to a tank pressurized with air or nitrogen to 80 psig. Pump the oil from the oil measuring tube into the tank. Failure to pump 1000 ml of oil into the tank within 5 minutes shall constitute failure of this test.

4.6.3.4 Refrigerant gage calibration. Using a mercury-manometer for the vacuum portion of the gage scale and a dead weight tester for the pressure portion of the scale, calibrate the refrigerant gages at each figured graduation. Nonconformance to 3.9.8 shall constitute failure of this test.

4.6.3.5 Shock. The service unit shall be tested for shock unpacked in accordance with MIL-STD-810, Method 516.1, Procedure II except the height of drop shall be 12 inches and the number of drops shall be a total of four, one on each of the four bottom corners, and Procedure VI. Nonconformance to 3.5.4 shall constitute failure of this test.

4.6.3.6 Low temperature storage. The service unit shall be tested for low temperature storage in accordance with MIL-STD-810, Method 502, Procedure I. The storage temperature shall be minus 65° F and the test duration shall be 6 hours. Nonconformance to 3.5.2 shall consittute failure of this test.

4.6.3.7 Humidity. The service unit shall be tested for humidity in accordance with MIL-STD-810, Method 507, Procedure I. Nonconformance to 3.5.3 shall constitute failure of this test.

4.6.3.8 Service unit check. The following tests shall be performed to check quality, performance, and operation of a service unit:

(a) In accordance with 4.6.3.1.1, except at only 200 microns Hg absolute pressure.
(b) In accordance with 4.6.3.1.2.
(c) In accordance with 4.6.3.1.3.
(d) In accordance with 4.6.3.1.4.
(e) In accordance with 4.6.3.2.1.
(f) In accordance with 4.6.3.2.4, except in 30-pound ± 2-pound increments.

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4.6.3.9 Human factors. The service unit shall be evaluated throughout testing to determine compliance with MIL-STD-1472A. Nonconformance to 3.7 shall consitute failure of this test.

4.6.3.10 Maintenance evaluation. Maintenance operations required during initial production tests shall be accomplished as specified in 3.8. Any assemblage or support equipment such as technical manuals, repair parts, and special tools provided under the contract for operation or conducting maintenance should be evaluated. Failure of the service unit to be maintained as specified in 3.8 shall constitute failure of this test.

4.6.3.11 Rain. The service unit shall be tested for rain in accordance with MIL-STD-810, Method 506, Procedure I. The enclosure shall be closed and fastened during this test and all four sides shall receive equal time exposure to the rain. Nonconformance to 3.5.5 shall constitute failure of this test.

4.6.3.12 High temperature. Conduct the test in accordance with MIL-STD-810, Method 501, Procedure I. The highest operating temperature (Step 4) shall be 120° F. Operate the unit during Step 5 as follows:

(a) In accordance with 4.6.3.1.1, except at only 500 microns Hg., absolute pressure.
(b) In accordance with 4.6.3.1.2.
(c) In accordance with 4.6.3.1.3.
(d) In accordance with 4.6.3.1.4 except for a period of 8 hours at the high temperature ambient.
(e) In accordance with 4.6.3.2.4.

Nonconformance to 3.5.1 or 3.5.2 shall constitute failure of this test.

4.6.3.13 Safety evaluation. The service unit shall be evaluated throughout testing to determine compliance with 3.6, and MIL-STD-882. Nonconformance to 3.6 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select service units at any time during the contract production period and subject these service units to the examination and tests marked "X" in Column 4 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected pre-production model and initial production service units. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison service unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison service unit(s) and shall be cause for refusal by the Government to continue acceptance of production service units until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on service units previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all service units accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Inspection of preparation for delivery.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

125.	Components not positioned and secured in the enclosure as specified for Level A or C.
126.	Method of preservation not as specified for Level A.
127.	Box not as specified for Level A.
128.	Strapping not as specified for Level A or B.
129.	Housing not as specified for Level B.
130.	Enclosure not secured as specified for Level C.
131.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

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5.1.1 Level A. The enclosure, containing all the components comprising the complete service unit, positioned and secured as provided for, shall be preserved in accordance with MIL-P-116, Method IID. The enclosure shall be considered the rigid metal container and the desiccant placed within.

5.1.2 Level C. The enclosure, containing all the components comprising the complete service unit, positioned and secured as provided for, shall be preserved and packaged in a manner to assure protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The enclosure, containing all the components of the service unit, preserved and packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, Grade B. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. The enclosure, containing all the components of the service unit, preserved and packaged as specified in 5.1, shall be covered with a close-fitting fiberboard housing consisting of two sides, two ends, and a top constructed in accordance with PPP-B-636, V3c, Style RSC. The housing shall extend to the bottom of the metal enclosure but above the skids and shall be secured in place with two straps. The straps shall conform to QQ-S-781, Type I or IV regular duty, Finish A, Size 3/8 inch by 0.015, seals as applicable, and shall be applied adjacent to and inside of each skid. Edge protectors shall be used on the top under the straps.

5.2.3 Level C. The enclosure, containing all the components of the service unit, preserved and packaged as specified in 5.1, shall be prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules. As a minimum the enclosure shall be secured to prevent pilferage of contents.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The service unit is intended for field servicing of refrigeration systems using R-114 as a refrigerant. The unit provides the following servicing functions:

(a) Refrigeration system evacuation, dehydration and leak test.
(b) Addition of measured volume of lubricant to the refrigeration system.
(c) Addition of measured weight of R-114 to the refrigeration system.
(d) Testing refrigeration system.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model and number required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(d) When initial production inspection is required, and the number of service units to be furnished (see 3.3).
(e) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production service units from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish service units conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and suppliers maintenance and operational manual to be furnished with each service unit.

6.5 Provisioning. The contracting officer should include provisioning requirements for such repair parts as necessary and instructions on shipment of service units.

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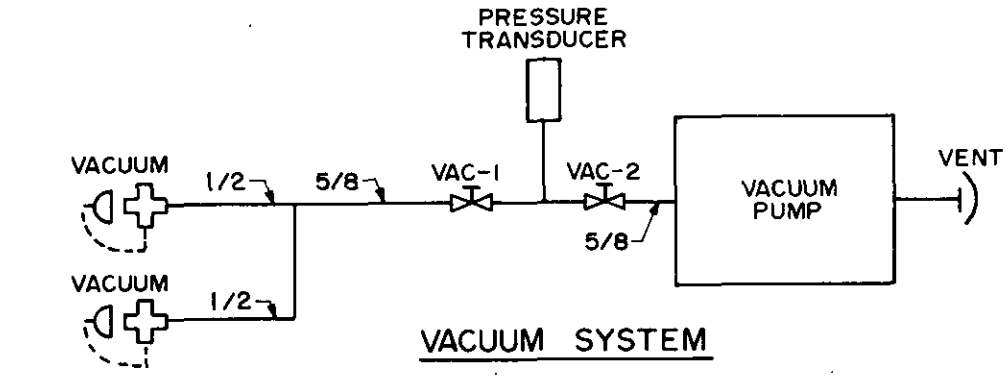
Army - ME

Preparing activity:

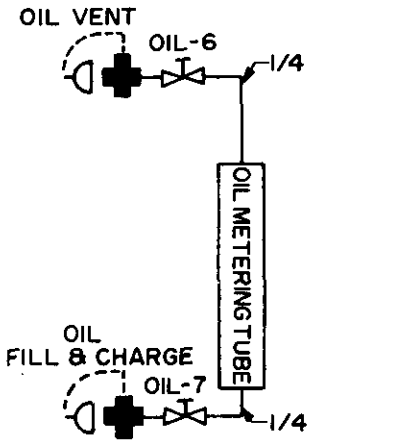
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Project No. 4130-A218

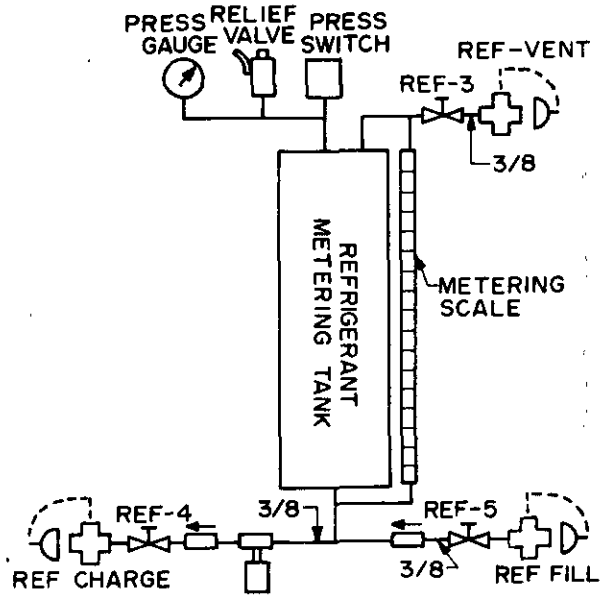
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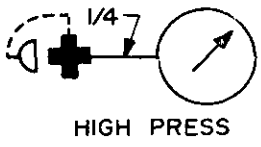
VACUUM SYSTEM

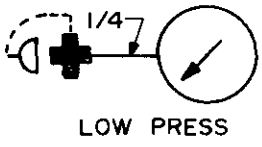


OIL CHARGING SYSTEM



REFRIGERANT CHARGING SYSTEM





REFRIGERANT GAUGES

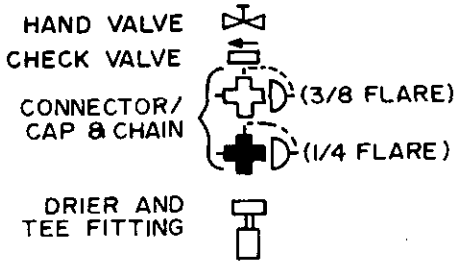


FIGURE I. SCHEMATIC DIAGRAM

X-2406B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an *envelope addressed to preparing activity.*

DOCUMENT IDENTIFIER AND TITLE

MIL-S-52744A(ME) Service Unit, Refrigeration System (MUST)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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MILITARY SPECIFICATION

SERVICE UNIT, REFRIGERATION SYSTEM (MUST)

This amendment forms a part of Military Specification MIL-S-52774A(ME), dated 11 December 1975, and is approved for use by the Mobility Equipment Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, under SPECIFICATIONS, FEDERAL, delete:

"QQ-S-698	- Steel, Sheet and Strip, Low Carbon.
WW-T-755	- Tube, Copper, Seamless, (for Refrigeration and General Use)."

2.1, under SPECIFICATIONS, MILITARY, delete:

"MIL-P-116	- Preservation-Packaging, Methods of."

and add:

"MIL-P-116	- Preservation, Method of."

PAGE 2

Under STANDARDS, FEDERAL, add:

"FED-STD-H28	- Screw-Thread Standards for Federal Services."

Under STANDARDS, MILITARY, add:

"MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1474	- Noise Limits for Army Materiel."

PAGE 3

2.2, under Other publications, delete in their entirety: "AMERICAN NATIONAL STANDARDS INSTITUTE, INC. ANSI)," "UNIFORM CLASSIFICATION COMMITTEE, AGENT," "NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT," AND "NATIONAL BUREAU OF STANDARDS."

and add:

"AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM).

B280 - Small Copper Tubing.

FSC 4130

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